



DEVELOPERS:

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FOODEXPRESS – FOOD ORDER & DELIVERY SYSTEM

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- i. Display all food items
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- i. Enter Order ID
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- i. Display menu items
- ii. Select item by number
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- ii. Include sales summary (orders, revenue)
- iii. Include customer satisfaction (reviews, average rating)

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- i. Back to Main Page

PROGRAMMING FUNDAMENTALS

PROJECT REPORT

Project Domain: FOOD ORDER AND DELEVERY SYSTEM

Development Team: ABDUL-RAHEEM, MUHAMMAD-OWAIS, MUHAMMAD-ATIF

Language: C++.

1. Executive Summary

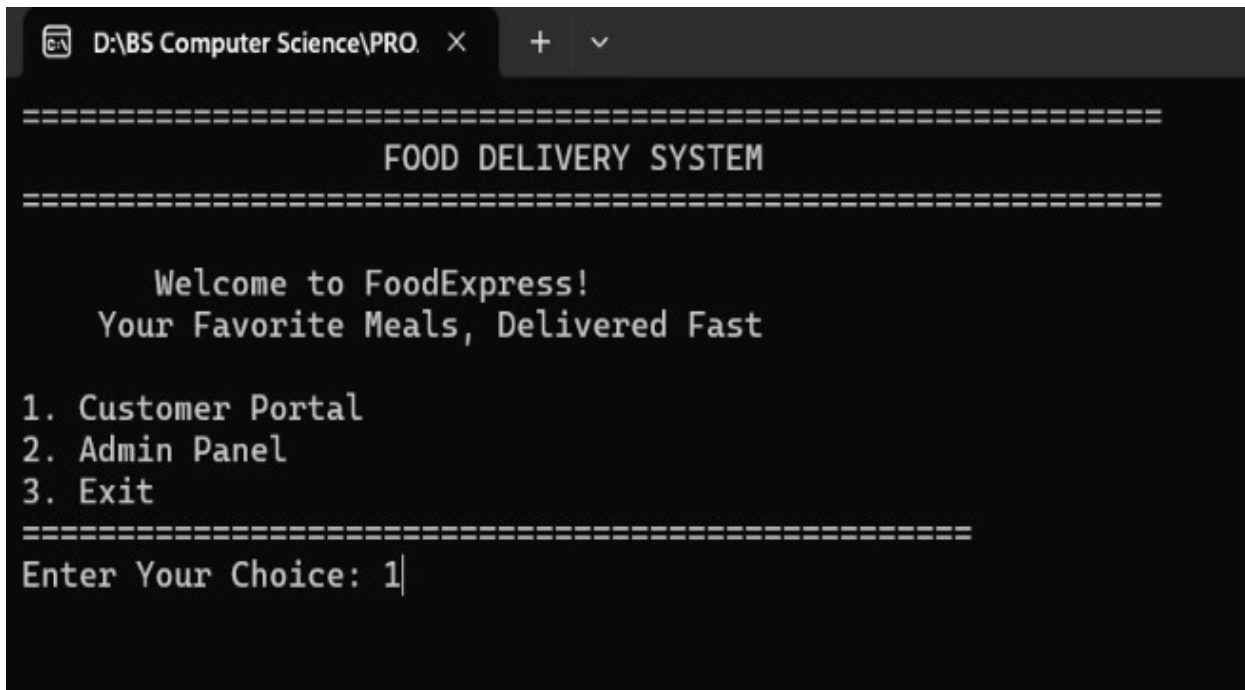
Food Express is a file-based Management Information System (MIS) developed to digitize and automate restaurant ordering and delivery operations. The system manages customers, orders, digital wallet payments, delivery logistics, and administrative analytics using a multi-tier architecture.

The goal is to replace manual restaurant processes with an efficient digital platform that ensures fast order processing, secure payments, accurate delivery tracking, and business intelligence reporting.

Engineering Division of Responsibilities

The development of the **FoodExpress – Food Ordering & Delivery Management Information System** was carried out collaboratively, with each team member assigned specialized engineering responsibilities across the system architecture.

Food Delivery system:

A screenshot of a C++ IDE window titled 'D:\BS Computer Science\PRO'. The main content is a text-based menu for a 'FOOD DELIVERY SYSTEM'. The menu is enclosed in a decorative border of equals signs. It displays a welcome message: 'Welcome to FoodExpress! Your Favorite Meals, Delivered Fast'. Below this, there is a numbered list of three options: '1. Customer Portal', '2. Admin Panel', and '3. Exit'. At the bottom, it prompts the user to 'Enter Your Choice: 1|', where the cursor is positioned after the number 1.

```
D:\BS Computer Science\PRO  X  +  v

=====
                        FOOD DELIVERY SYSTEM
=====

Welcome to FoodExpress!
Your Favorite Meals, Delivered Fast

1. Customer Portal
2. Admin Panel
3. Exit
=====
Enter Your Choice: 1|
```

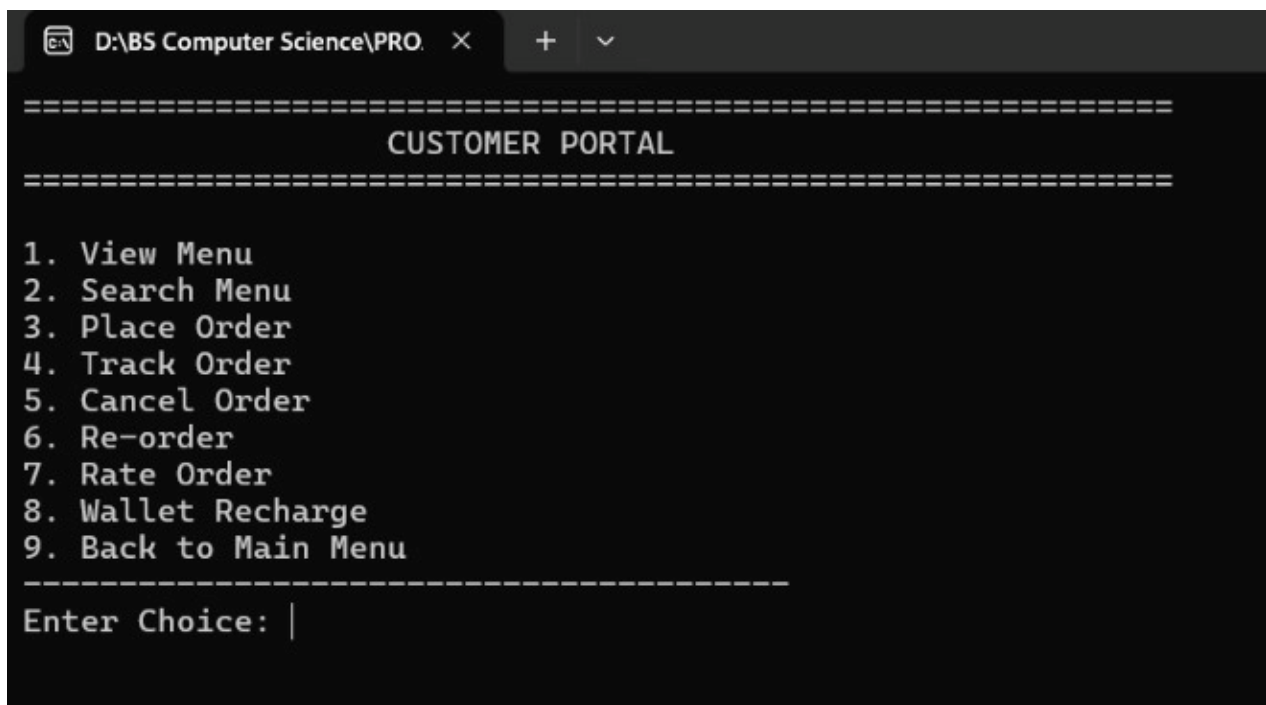
Explanation:

Working of the Program

1. An integer variable `ch` is declared to store the user's menu choice.
2. A do-while loop is used so that the menu is displayed **at least once** and continues to appear until the user selects the exit option.
3. Inside the loop:
 - The function `showHeader("FOOD DELIVERY SYSTEM")` displays the title of the system.
 - Welcome messages are printed on the screen.
 - Three menu options are shown:
 - Customer Portal
 - Admin Panel
 - Exit
 - The function `printLine('-', 50)` draws a decorative line to improve the user interface.
 - The user is asked to enter a choice, which is stored in the variable `ch`.
4. A switch statement processes the user's choice:
 - **Case 1:** Calls the `customerMenu()` function to open the customer portal.

- **Case 2:** Calls the `adminMenu()` function to open the admin panel.
 - **Case 3:** Displays a thank-you message and exits the loop.
 - **Default:** Displays an error message for an invalid choice and pauses the screen.
5. The loop continues running until the user enters **3**, which ends the program.
 6. Finally, return 0; indicates successful program termination

Customer Portal:



```
D:\BS Computer Science\PRO. x + v

=====
                        CUSTOMER PORTAL
=====

1. View Menu
2. Search Menu
3. Place Order
4. Track Order
5. Cancel Order
6. Re-order
7. Rate Order
8. Wallet Recharge
9. Back to Main Menu
-----
Enter Choice: |
```

Explanation:

1. An integer variable `ch` is declared to store the customer's menu choice.
2. A do-while loop is used to repeatedly display the customer menu until the user chooses to return to the main menu.
3. Inside the loop:
 - The function `showHeader("CUSTOMER PORTAL")` displays the title of the customer section.
 - Several menu options are displayed for the customer:
 - View Menu, Search Menu, Place Order, Track Order, Cancel Order,
 - Rate Order, Wallet Recharge, Back to Main Menu

- The function `printLine('-', 40)` prints a horizontal line to improve the visual layout.
- The user is prompted to enter a choice, which is stored in the variable `ch`.

Place New order:

```
D:\BS Computer Science\PRO  x  +  v

=====
                        PLACE NEW ORDER
=====

--- Customer Details ---
-----
Name: Muhammad Owais
Phone (11 digits): 03330356589
Delivery Address: Gulberg

--- Order Items ---

-----
No.  Item                Category      Price      Status
-----
1    Zinger Burger        Fast Food    Rs 350      Available
2    Chicken Pizza        Fast Food    Rs 899      Available
3    Fries (Large)         Sides       Rs 200      Available
4    Pasta Alfredo         Italian      Rs 450      Available
5    Cold Drink            Beverages    Rs 120      Available
6    Chicken Biryani       Desi        Rs 380      Available
7    Ice Cream             Dessert     Rs 250      Available
8    Caesar Salad          Healthy     Rs 280      Available
9    Shawarma              Fast Food    Rs 320      Available
10   Chicken Karahi        Desi        Rs 1200     Unavailable
11   Brownies              Dessert     Rs 180      Available
12   Fresh Juice           Beverages    Rs 150      Available
-----

Select item (1-12): |
```

Explanation:

Customer Information

At the start, the function displays the heading “**PLACE NEW ORDER**”. Variables are declared to store customer information, order details, and billing data. A unique order ID is generated using the `generateOrderID()` function.

The program asks the customer to enter their name, phone number, and delivery address. The phone number is validated using the `validatePhone()` function to ensure it contains exactly 11 digits. The input continues until a valid phone number is entered.

Item Selection Process

After entering customer details, the customer is allowed to select food items. The food menu is displayed using the **showFoodMenu()** function. The customer selects an item number and quantity.

The program checks:

- If the selected item number is valid
- If the selected item is available
- If the entered quantity is greater than zero

If any invalid input is detected, an error message is shown and the user is asked again.

Each selected item's total price is calculated and added to the subtotal. The item details are also saved in a string to keep a record of ordered items. The user can continue adding items until they choose not to add more.

Order Validation

After item selection, the program checks whether at least one valid item was added to the cart. If no item was added, the order is cancelled to avoid empty orders.

Charges Calculation

The delivery fee and estimated delivery time are calculated based on the delivery address. Tax is calculated using a fixed tax rate. All these charges are later used to generate the final bill.

Promo Code Handling

The user is asked if they have a promo code. If a promo code is entered, it is validated using the `validatePromoCode()` function. A valid promo code applies a discount to the order, while an invalid code displays an error message.

Tracking order:

```
D:\BS Computer Science\PRO. x + v

=====
                        TRACK YOUR ORDER
=====

Enter Order ID: 1001

--- Order Tracking ---
-----
Order ID: #1001
Placed on: 29/12/2025 17:58
Customer: Muhammad Owais
Delivery Address: Gulberg
-----

--- Order Status ---
[DONE] Order Confirmed
[CURRENT] Preparing...
[PENDING] Out for Delivery
[PENDING] Delivered

Rider: Ahmed Ali (ID: 201)
Estimated Time: 25 minutes
Total Amount: Rs 8746
Payment: Cash
-----

Press Enter to continue...
```

Explanation:

Order ID Input

The function starts by displaying the heading “TRACK YOUR ORDER”. The user is asked to enter their order ID, which is stored in the variable orderID.

File Handling

The program opens the orders file using ifstream. If the file cannot be opened or no orders exist, an error message is shown and the function ends safely.

Reading Order Data

The program reads the file **line by line**. Each line represents one order. The line is split using a string stream, where data fields are separated by commas. These values are stored in an array for easy access.

If the data format is incomplete, the program skips that line to avoid errors.

Order Matching

The entered order ID is compared with the order ID stored in the file. When a match is found, the program displays detailed order information such as order ID, order date, customer name, and delivery address.

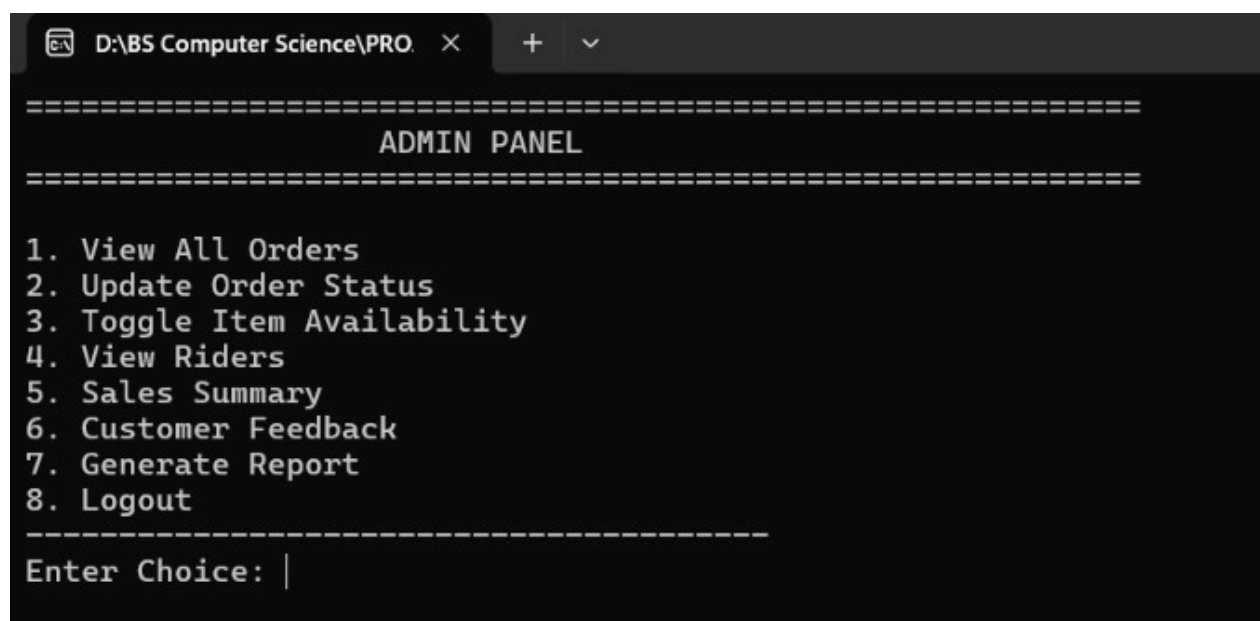
Order Status Tracking

The order status is retrieved from the stored data. Based on the status value (Pending, Preparing, Out for Delivery, or Delivered), the program displays a **step-by-step progress** of the order. Completed steps are marked as done, the current step is highlighted, and remaining steps are shown as pending.

Rider and Payment Details

The rider's name is fetched using the rider ID and displayed along with estimated delivery time. The total bill amount and payment method are also shown for customer reference.

Admin Panel:



```
D:\BS Computer Science\PRO. x + v
=====
                        ADMIN PANEL
=====

1. View All Orders
2. Update Order Status
3. Toggle Item Availability
4. View Riders
5. Sales Summary
6. Customer Feedback
7. Generate Report
8. Logout
-----
Enter Choice: |
```

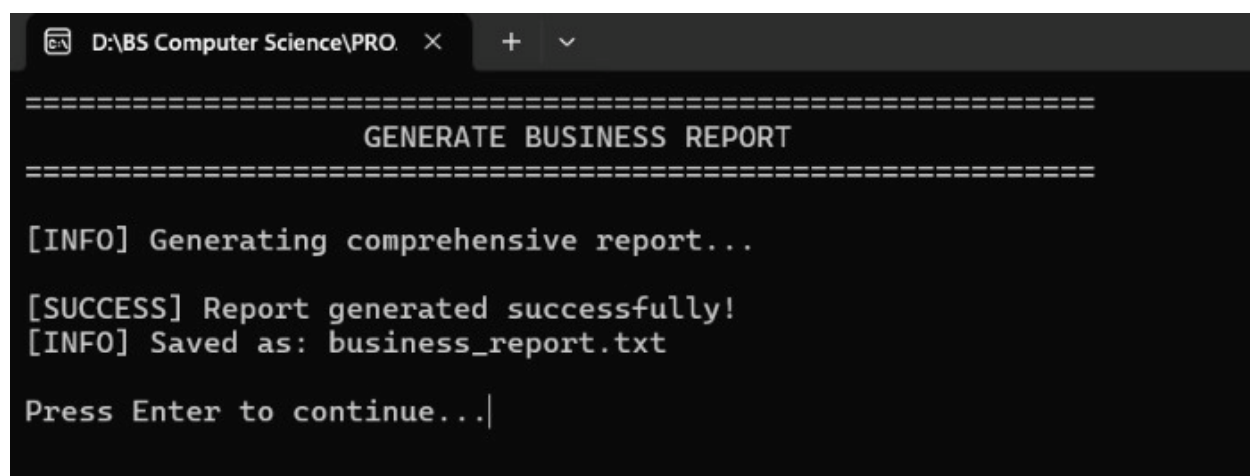
Explanation:

The view Orders() function allows the **admin or staff** to view **all orders placed in the system** in a **well-formatted table**. It opens the orders file and checks if it exists. If the file is missing or empty, an error message is displayed and the function exits.

The function prints a table header showing order ID, customer name, phone number, total amount, status, rider ID, and date/time. Each line in the file is read and split into separate fields using commas. Incomplete lines are skipped to avoid errors. Valid order details are displayed row by row with proper alignment using formatted output.

Finally, the file is closed and the program pauses, allowing the admin to view all orders clearly. This function helps in **efficiently monitoring and managing all orders** in the system.

Function for generate Business report:



```
D:\BS Computer Science\PRO. x + v
=====
                        GENERATE BUSINESS REPORT
=====

[INFO] Generating comprehensive report...

[SUCCESS] Report generated successfully!
[INFO] Saved as: business_report.txt

Press Enter to continue...|
```

Explanation:

The generate Report() function generates a **comprehensive business report** for the Food Delivery System. It creates a text file named business_report.txt containing sales and customer satisfaction details.

The function starts by displaying a heading “**GENERATE BUSINESS REPORT**” and a message indicating that the report is being generated. It then opens the orders file and reads each order line by line. For each valid order, it increments the **total orders count** and adds the order amount to calculate **total revenue**.

Next, the report includes a **sales summary** showing the total number of orders and total revenue earned.

The function then opens the feedback file and reads each feedback entry to calculate **total reviews** and **average customer rating**. These statistics are added under the **customer satisfaction** section of the report.

Finally, the report is saved with a heading, generation date/time, and end-of-report markers. A success message is displayed on the screen indicating that the report has been generated and saved.

This function helps the admin **analyse business performance, track revenue, and monitor customer satisfaction** effectively.

Conclusion:

The *Food Express – Food Ordering & Delivery System* successfully converts manual restaurant work into a simple digital system. It allows customers to place, track, and manage orders easily, while the admin can view all orders and generate business reports.

The project uses core C++ concepts like menus, functions, file handling, and input validation, making it reliable and user-friendly. Overall, this system meets its goals and provides a strong base for future improvements such as adding a database or a graphical interface.